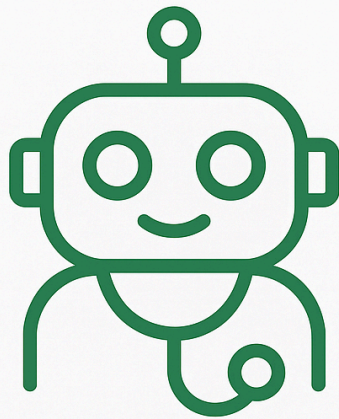


CSE327

Software Engineering

Project Proposal



Dr. Chatbot

Group Information:

Sl. no	Name	ID
1.	Sadia Eva	2231138042
2.	Zaima Hossain	2231013042
3.	Shafaat Bin Zaman	2232015042

Project Description

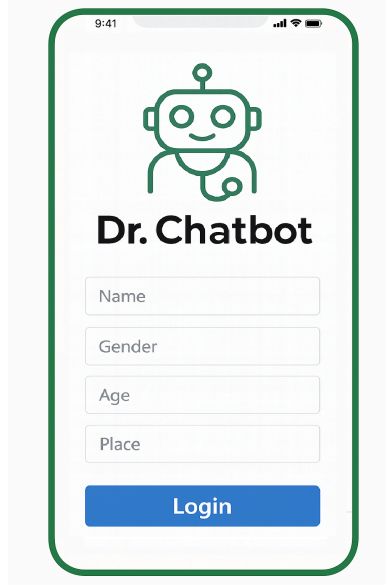
The Health Symptom Checker app is mobile-friendly and allows users to submit their health symptoms in natural language. Using NLP techniques and a Hugging Face LLM, the backend system analyses the input, matches the symptoms mentioned with those of prevalent diseases (such as gastritis, fever, flu, dengue etc.), and then intelligently delivers advice. This involves foreseeing potential causes, offering quick remedies or preventative measures, and emphasising "red flag" symptoms that might require an immediate evaluation by a doctor.

The goal of this project is to create and run a system that helps people to understand their health problems quickly and accurately. The app is a helpful first step in the decision-making process, even though it cannot replace professional medical treatment, especially in places where there is a doctor shortage or when people are unwilling to seek urgent medical care.

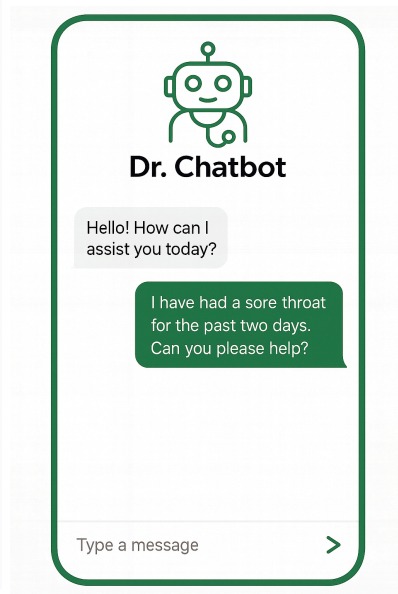
Features

1. **Symptom Input:** Users can type symptoms and the model could comprehend natural human text.
2. **Disease Prediction:** AI model predicts the most likely conditions with confidence scores.
3. **Chat-Based Interface:** Users will feel like talking to a health assistant.
4. **Red-Flag Detection:** Automatic alerts for critical symptoms requiring urgent medical care.
5. **Basic Medical Advice:** Short recommendations (such as hydration, rest, consult a doctor, etc.).
6. **Multilingual Support (Optional) :** Interface and NLP support for English and Bangla.
7. **User History Tracking(Optional):** Allow users to review past records and recommendations.

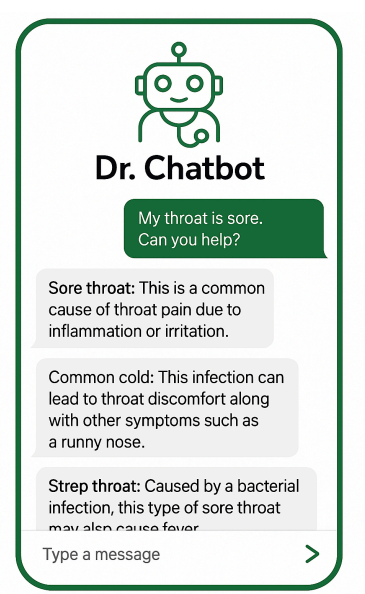
UX Diagram



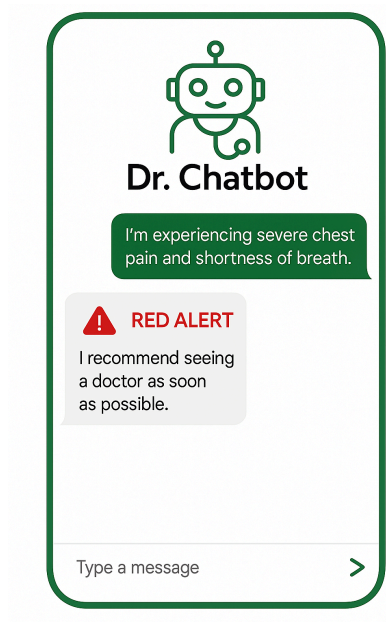
Login Screen



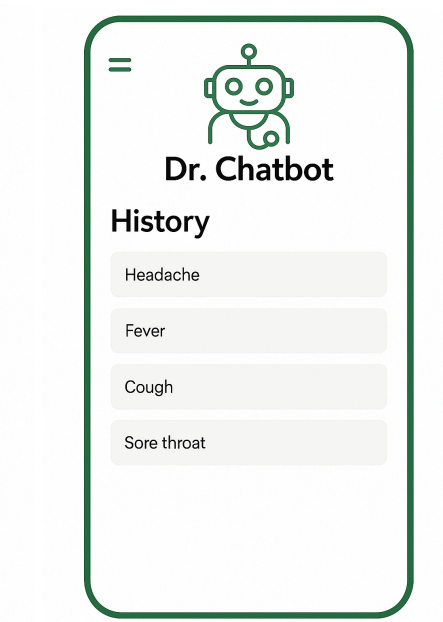
Conversation Interface



Prediction Screen



Red Alert Screen



History Page

Framework

The tools we are planning to use:

- **Frontend- Flutter**
- **Backend- Flask**
- **NLP, LLM Model- Hugging Face**
- **Database- Firebase**
- **Testing- Android Emulator**

Weekly Breakdown of Project Plan

Week 1 : Plan the project and research.

Week 2 : Collect data and preprocess and explore symptom datasets.

Week 3 : Select the model and experiment whether it shows accuracy or not.

Week 4 : Implement text preprocessing (tokenization, NER, classification).

Week 5 : Interpret symptoms and generate response as output.

Week 6 : Design UI for android.

Week 7 : Integrate frontend and backend.

Week 8 : Enhance with optional features and bug testing.

Week 9 : Take sample user inputs, analyze accuracy and show model performance.

Week 10 : Prepare final report and presentation, user manual and demo.